

achieved in this aspect, and so the focus is shifting to low cost, short time and ease of testing.

"The drive to reduce the cost of test is forcing changes in the way instruments communicate and the way they are programmed. As examples, look at the rise in the number of LAN eXtended for instrumentation (LXI) instruments (over 1500), graphical programming and the acceptance of modular tests such as PCI eXtensions for Instrumentation (PXI)," says Bob Stasonis, Pickering Interfaces, USA.

A lot of effort is also focused on making the test engineer's job of routing stimulus or measurement instruments to and from the unit-under-test easier. "To that end, we have focused on switching the density and bandwidth. As products become more complex, more inputs/outputs (I/Os) need to be managed. Also, between new high-speed serial buses and wireless products, the need for even higher number of switching points and getting a few more GHz where possible is the priority," remarks Stasonis.

On the technology front, the growth of wireless technologies has been driving a lot of innovations in the T&M world. Higher signal bandwidths and increasingly complex modulation schemes, strict QoS requirements, need for real-time monitoring across networks, etc are driving the need for faster and more capable instruments.

The recent multicore mania, surge in digital broadcasting, more stringent quality norms across industries, convergence of design and development activities across the product lifecycle, need to shorten the time-to-market like never

Software development coupled with improvements in areas like LED lighting allows vision inspection systems to improve test speeds and reduce marginal errors



Automatic test equipment for avionics (Courtesy: ST Aerospace)

before, and a lot of other new technological and market trends are also triggering various developments in the T&M world. In the rest of this article, let us delve into some of these recent developments and trends in test and measurement.

Great improvements in vision test systems

When we think of test systems, machine vision is probably the first thing that comes to our mind. Machine vision involves the capture and analysis of images to monitor and control an activity. Generally, it is used for automatic inspection and process control in assembly lines, guidance of robots, etc. The light source used is a very important factor in vision test systems because it affects how accurately the image is captured—and hence how accurately defects can be spotted.

Of late, there have been significant improvements in machine vision systems with the use of light-emitting diodes (LEDs) for the light source. LEDs offer many benefits like high efficiencies, reproduction of varied colours and longer lifetimes. Moreover, the last few years have seen significant improvements in LED technology in terms of brightness, efficiency, thermal stability and optics—all of which are now being harnessed by machine vision companies

to improve their test systems.

We also see test systems with increased strobe speeds emerging to meet new machine vision applications. Cameras capable of 1000 frames per second (fps) are becoming the standard. These take hundreds or thousands of inspections on a part looking for a pass condition, as against the earlier norm of one or a few images. Software is also becoming smart enough to process and analyse these images swiftly and accurately.

Stasonis confirms, "Improved lighting and, most importantly, software algorithms have improved vision test systems. Software development coupled with improvements in areas like LED lighting allows vision inspection systems to improve test speeds and reduce marginal errors."

Emergence of multicore systems

Multicore processors are fast becoming popular as these help achieve performance gains without increased clock rates. Of course, the world of T&M has to keep pace with developments across the technological spectrum. Hence test engineers are now beginning to use multicore processors to develop automated test applications capable of achieving the highest-